



Slicer Project

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Introduction

The objective of this project is to dive into the practical application of **3D Slicer**, an open-source platform that has become a standard for medical image informatics and visualization. Rather than just following a tutorial, this work is designed as an **exploratory journey** through the process of transforming raw **DICOM data** into meaningful 3D representations.

Throughout the development of these tasks, we focused on learning the core concepts of volumetric processing, from the initial **segmentation** of anatomical structures to the creation of advanced **3D visualizations**. This report serves as a log of our progress, where we explain the methodology used to solve each exercise and, importantly, discuss the **troubles and challenges** we encountered along the way—such as refining complex geometries or navigating the vast ecosystem of **Slicer plugins**.

The project is structured into the following key areas:

- **Segmentation and AI Exploration:** We compare manual segmentation techniques with **AI-based modules** to understand the current capabilities and limitations of automated medical tools.
- **Visualization and Metrics:** We explore how different **transfer functions** and **markups** can be used to extract quantitative insights that would be useful for a medical professional.
- **Plugin Integration:** We investigate advanced tools not covered in our standard lab sessions to see how specialized plugins can tailor the software to specific clinical tasks.

Ultimately, this project is about gaining hands-on experience and a deeper understanding of how to bridge the gap between raw medical data and **meaningful 3D insights**.

1 Segmentation task

The last number of the Enric's ID card is the 46. Since the division of 46 by 6 has a remainder of 4, it will be used the fourth option of data that is a segmentation of the lungs along with the major thoracic bones. To do it the data will be downloaded from the Download sample data option in the Welcome to slicer module. The data choose is the CTCchest.

The first step to segmentate the pulmons is to add a segment for it. It is called "Pulmons" and it has been selected as a dark red color that usually is used for the organs of the body. The segmentation of the organ is done by different modes of segmentation:

1.1 Threshold segmentation

This is the most basic and most widely used segmentation technique in medical images. The technique is based on the set of a threshold value for the intensity of each pixel in the image. All those pixels that are in the range of the threshold value will be shown at the same time that all the pixels that are not in the range will not be shown. It is a kind of binari tool to differentiate types of pixels.

To set the threshold value in the slicer program it is used a tool that allows you to select the organ's zone that you want to see by a cerquel in the image. In **Fig. 1**. it can be seen the usage of the selection tool to select a part of the pulmons.

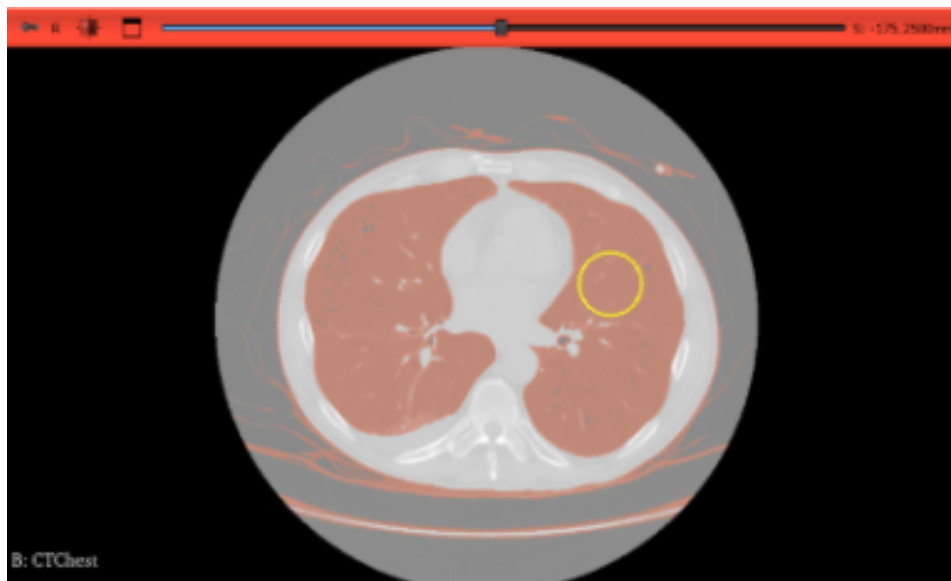


Figure 1: Visualitization of the selected zone of the lungs to set the threshold value.

With this tool it has been set a threshold range between the values of $(-931, -343)$ that all the pixels of the lung share. The 3D visualization using just this technique is the one that is shown in the **Fig. 1**.

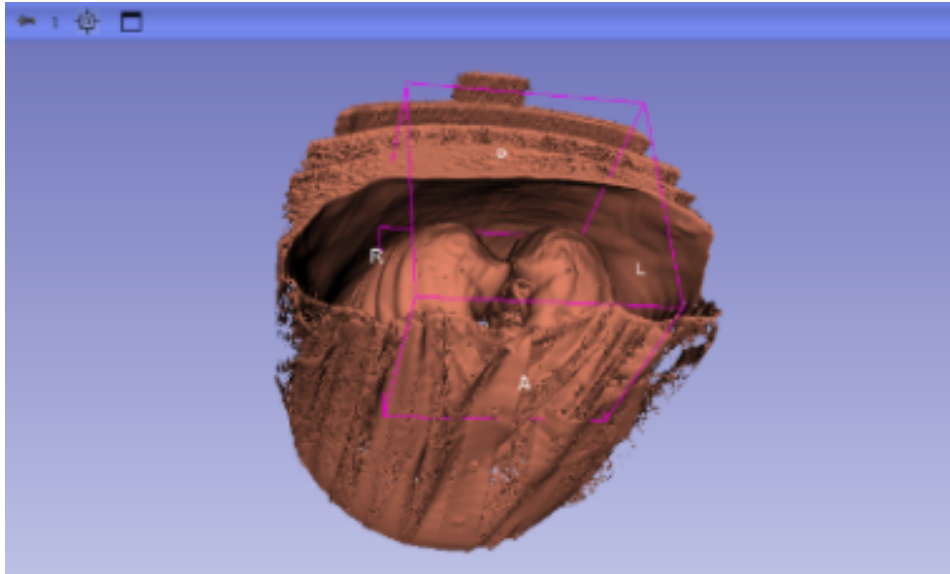


Figure 2: visualization of the result of the threshold segmentation tool on the CTchest model.

By looking at the results of the first segmentation it is clear that the pulmons pixels are in the threshold range but also a lot of non important body data is on that range so it needs to be eliminated from the visualitization to see clearly the organ. To do it there are other segmentation techniques that are useful.

1.2 Scissors segmentation

The segmentation scissors tool is a good technique to clean our image from undesired data. By using the scissors tool that is on the left part of the program screen can be selected the wrong data of the image by circling it in a plane, as it is seen in *Fig. 3*.

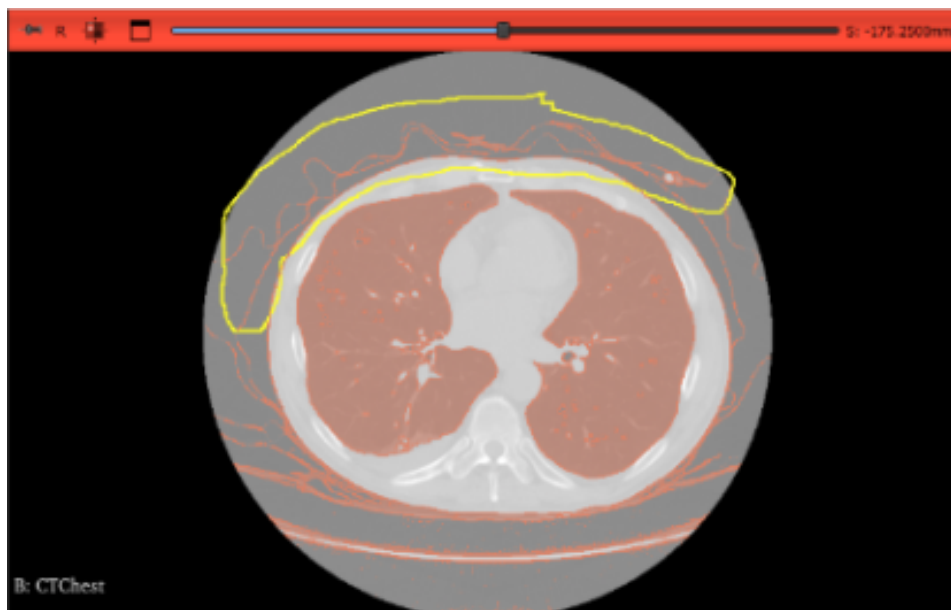


Figure 3: Usage of the scissors segmentation tool on the upper vision of the lungs to eliminate data that is not from our interest.

This technique allows us to erase all the undesired data of the image, in the **Fig. 3** can be seen our model once this technique is used on it.

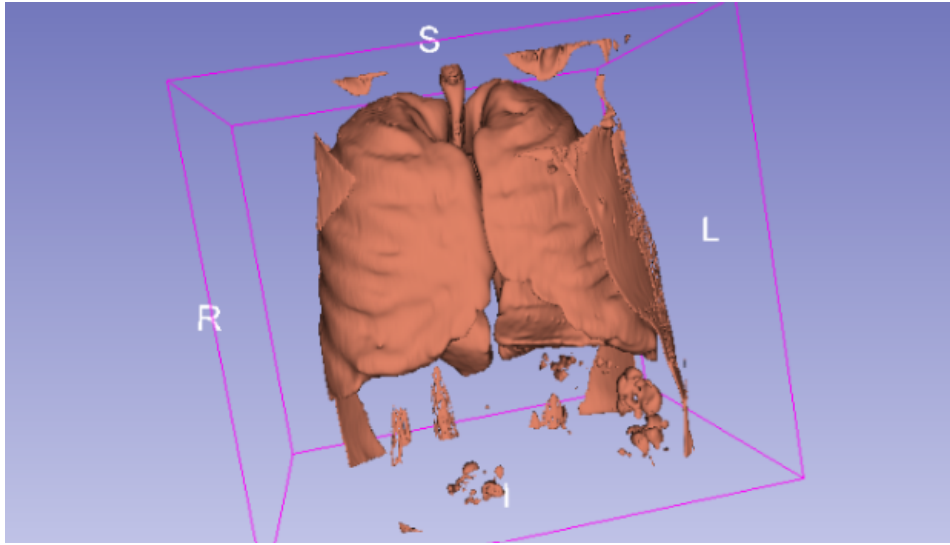


Figure 4: Results of the scissor segmentation tool usage on the CTChest mode.

1.3 Islands segmentation

Another technique to edit the segmentation is the called islands. In the left column of the slicer program can be selected the island tool and by selecting the rest of the pixels that are not from the lungs they can be erased from the image as it is shown in the **Fig. 5**.

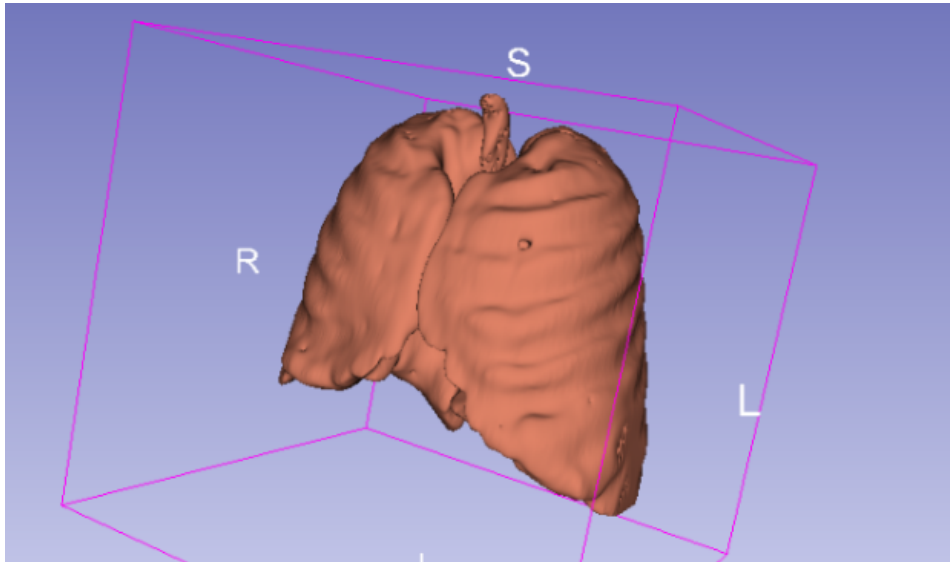


Figure 5: Visual result of island tool usage on the CTChest model.

The objective of the exercise is to visualize the lungs and the thoracic bones so to do it another segmentation is created called “Major thoracic bones”. Another threshold range is created with the values of (85, 463). The results are shown in the **Fig. 6**.

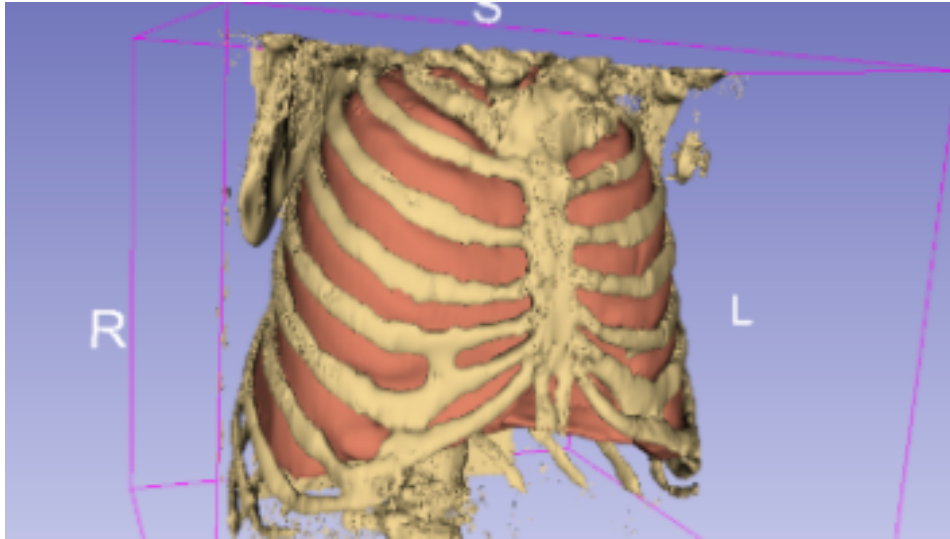


Figure 6: Visual result of the thoracic bones segmentation on the CTchest model.

By using the scissors and the islands techniques explained before the image is processed to discard the visualization of the pixels that are not interesting for this visualization. The result is shown in the *Fig. 7*.

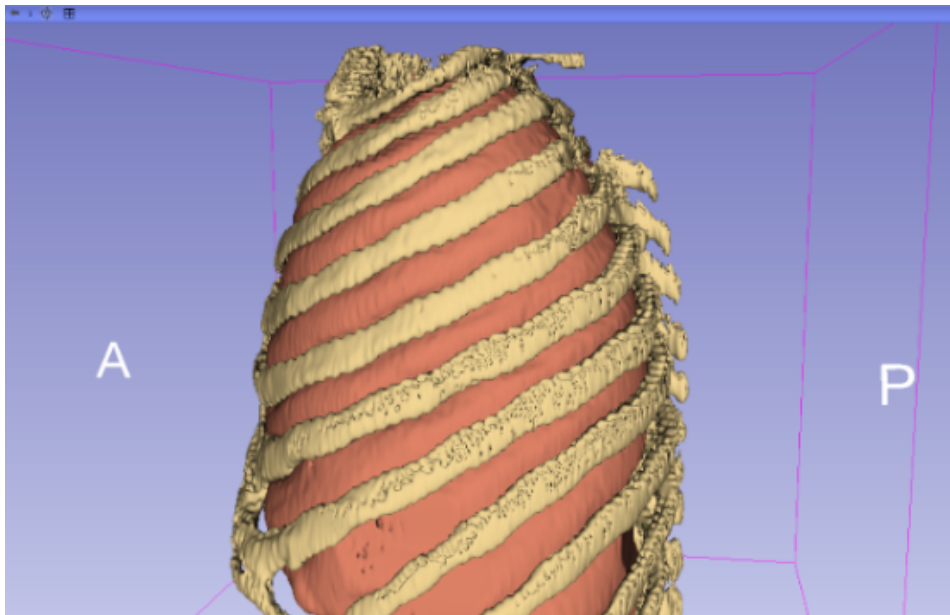


Figure 7: Lateral vision of the scissor tool usage on the CTchest model to eliminate the bones parts are not from the torcic bones. This is the final segmentation result of the exercise.

2 Visualization task

Using the same volume dataset: a. Choosing two different presets for the transfer function, produce different changes into the transfer function to obtain different results.

Using the volume rendering module, we can use different presets. Those presets have different transfer functions that produce very different outputs. For this exercise we have selected the following presets:

- CT-Bones
- CT-Lungs

2.1 CT-AAA

This preset is for visualizing the thoracic bones. We modified slightly the transfer function to make it look as this:

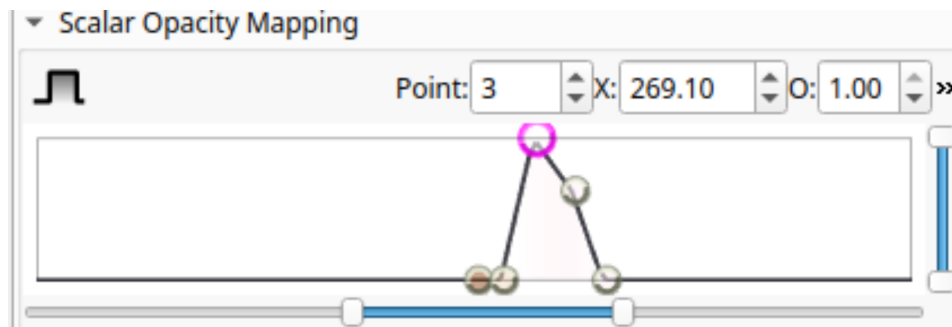


Figure 8: Visualization of the thoracic bones using the CT-Bones preset with a modified transfer function.

Which produced the following output:

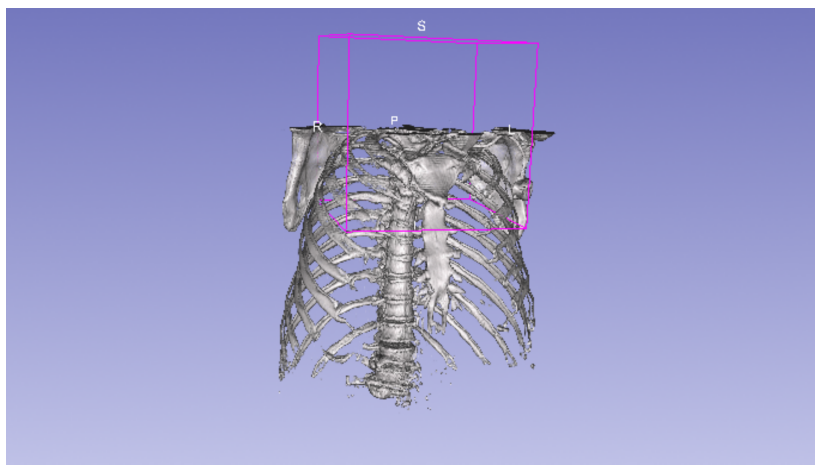


Figure 9: Output of the CT-Bones preset with a modified transfer function.

2.2 CT-Lungs

For the lungs preset, we leaved it as it was, because it was selecting correctly the voxels corresponding to the lungs.

Transfer Function:

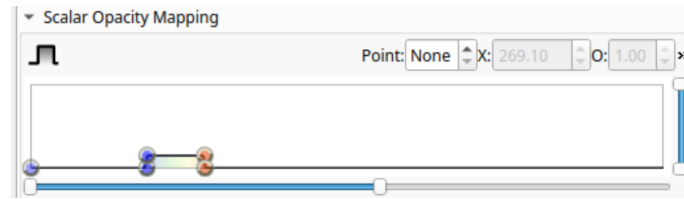


Figure 10: Transfer function of the CT-Lungs preset.

Output:

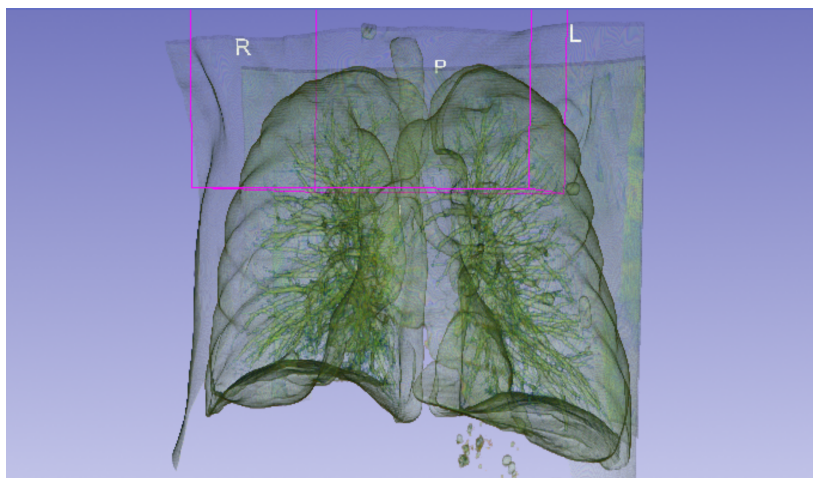


Figure 11: Output of the CT-Lungs preset.

3 Measurement task

Define a set of measurements or markups that would enable a hypothetical medical expert to gain insights into the volumetric model used in Task 1

For the lungs segmentation. The following markups could be of interest in order to monitor lung progression for smokers along time.

- **Lung Perimeter:** Measure the total volume of each lung to monitor changes over time, which could indicate disease progression or improvement.
- **Nodule Height:** Place fiducial markers on any detected nodules within the lungs to measure their size and track growth or shrinkage over time.
- **Airway Width:** Use line markups to measure the diameter of major airways (e.g., trachea, bronchi) to assess for any narrowing or obstruction.

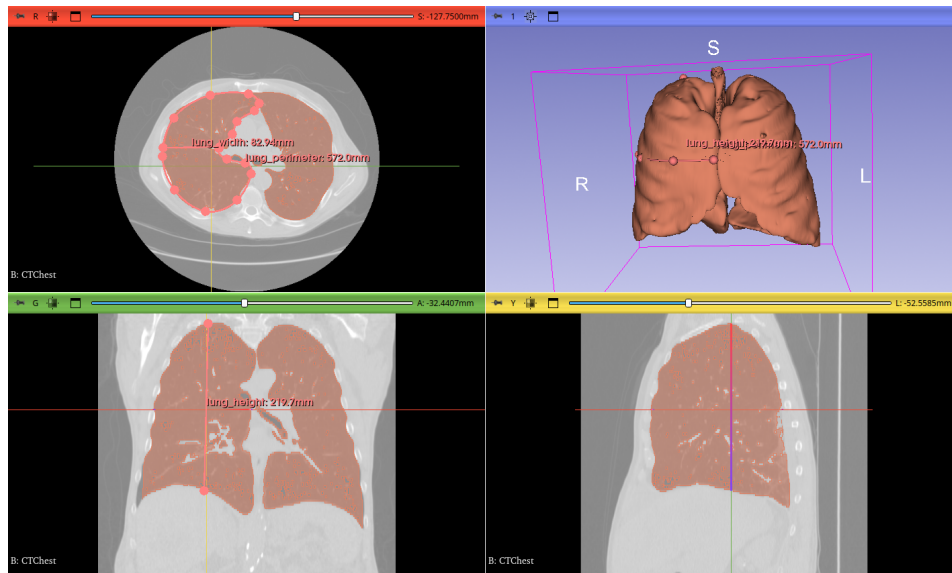


Figure 12: Visualization of the lungs segmentation with fiducial markers on detected nodules and line markups on airways.

Those metrics have been obtained with the markups module. The following results were obtained:

Measurement	Value
Lung Perimeter	572.0 mm
Lung Height	219.7 mm
Lung Width	82.9 mm

Table 1: Lung measurements obtained from the segmentation model.

4 Report Task

Combine, within a single image, a rendered portion of the volumetric dataset together with the anatomical 2D views, the various measurements or markups generated in the previous task and the models generated from the segmentation task. Create meaningful visualizations that would enable a hypothetical medical expert to gain insights into the key regions of interest within the examined volumetric model. Imagine that these visualizations (snapshots) will be included as part of a report prepared for such an expert.

Using the segmentation model generated in task 1, we generated meaningful visualizations of the model and the markups. This capture enables the doctor to clearly analyze the structure and shape of the lungs, as well as seeing the various lung measurements

As we already had the segmentation and the markups done, the process was very straightforward.

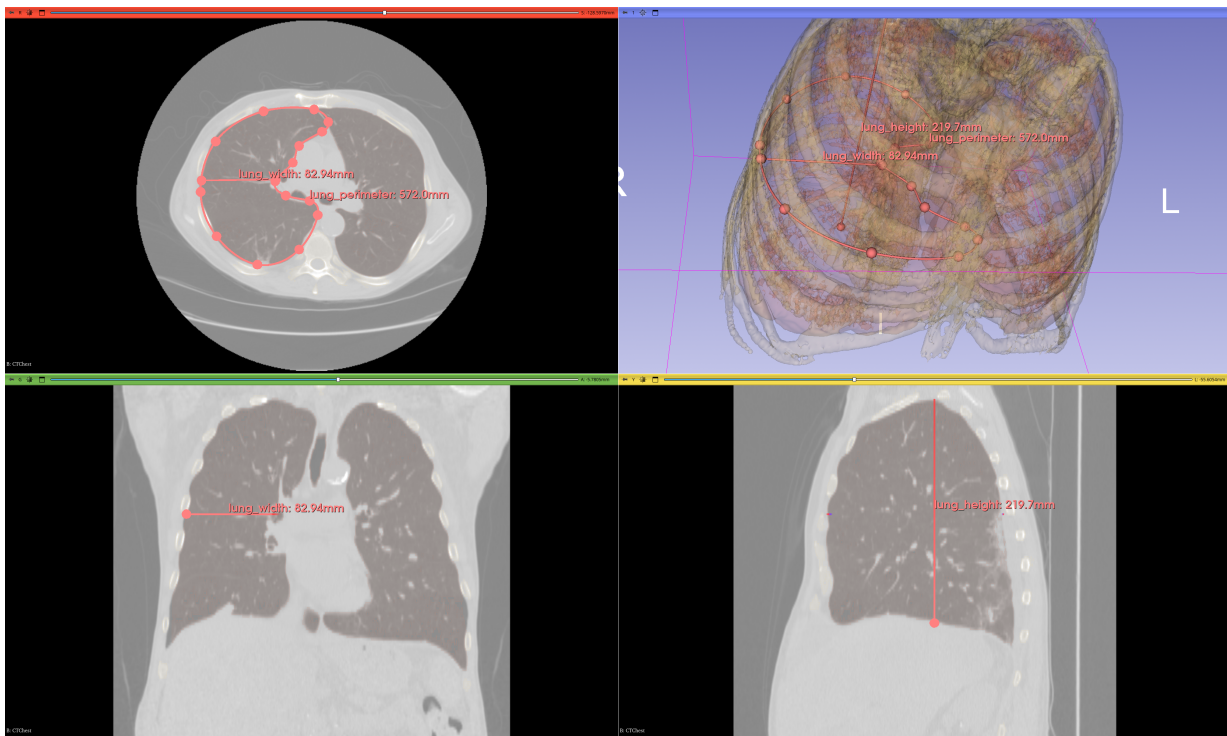


Figure 13: Combined visualization of the lungs segmentation, anatomical 2D views, and measurements.

In order to enhance the visualization of the perimeter markups, the opacity of the segmentation model was reduced, leading thus, a clearer view.

5 AI segmentation task

Test some of the AI segmentation plugins to segment the same anatomical structures:

Using the newest data processment tools has been created plugins to work with models. Concretely the artificial intelligence has been used to develop those plugins. A good example is the plugin that will be used in this exercise: Total Segmentator. Those plugins are created and shared by medical engineers from universitys, for example the one that is used has been created by the Research and Analysis team of the Research Hospital in Basel. The plugin consists on a tool for segmentation of most major anatomical structure in any CT or MR image. The tool was trained on a wide range of different CT and MR (different scanners, institutions, protocols...) and therefore works well on most images.

The steps to use the plugin are the following:

5.1

Enter to the Total Segmentator module and select the input button as can be seen in the *Fig. 14*

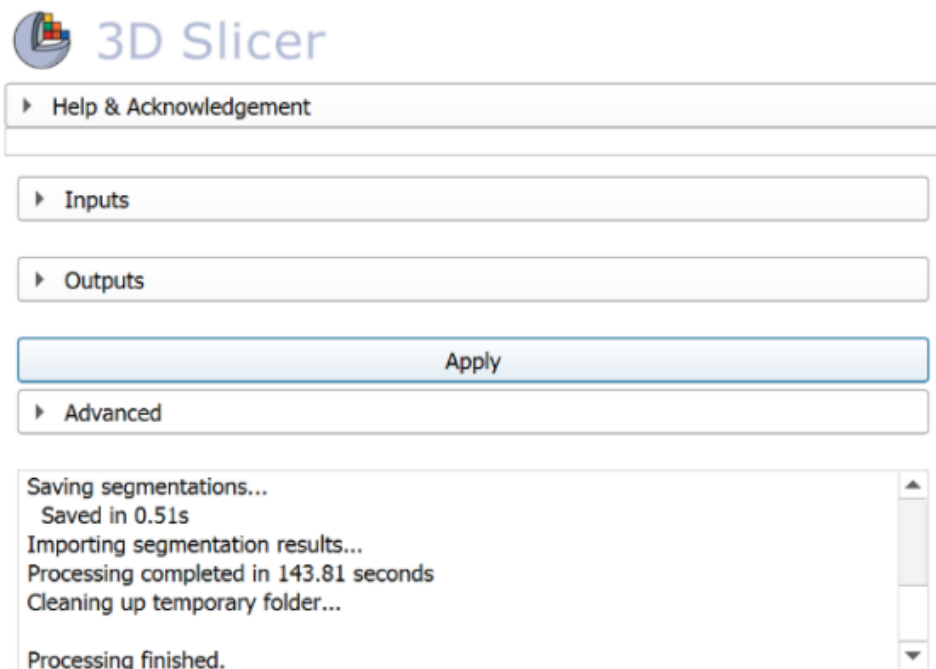


Figure 14: Initial menu of the Total Segmentation plugin tool.

5.2

In the input part of the menu will be necessary to select the 3D model which is used, in this case the CTChest as it is visible in the input volume. It is necessary also to select the segmentation task that is required, it is possible to to segmentate different parts of the data alone or all the parts at the same time. As it is not possible to segmentate the lungs and the thoracic bones at the same time it has been chosen to segmentate all the parts. This plugin can work with extreme precision since is using a powerful tool of segmentation. The drawback of it is that it requires a high performance laptop to run it and approximately one hour of computer work. Since the computer used to do it is

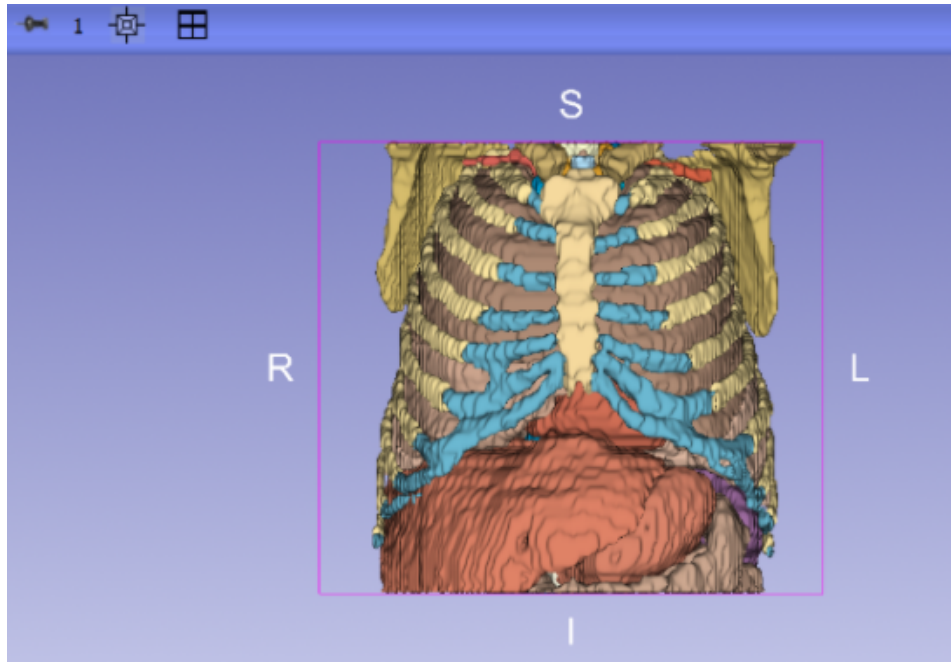


Figure 16: First result of the usage of the Total Segmentation plugin tool in the CTChest data.

not powerful enough, the Fast option has been selected. This option runs a fast version of the plugin without the level of detail that can perform but in a really lower time. The computer used lasted 10 minutes to obtain as a result the image of the *Fig. 16*.

▼ Inputs	
Input volume:	CTChest ▼
Segmentation task:	total ▼
Fast:	☒

Figure 15: Option menu of the Total Segmentation plugin tool.

Finally by selecting the visualize or not visualize option in the segment editor for the segments that we are not interested in as it is visible in the Fig 15, it is possible to achieve the final segmentation of the lung and the thoracic bones with a regular precision as it is visible in the Fig 16.

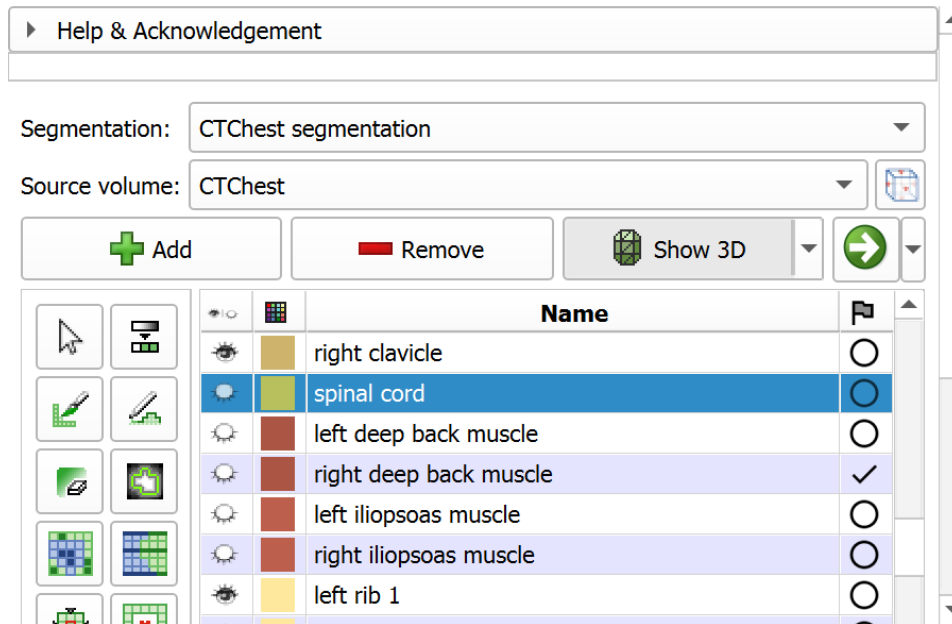


Figure 17: List of segmentation created by the IA tool and selection of the visualization option for the segments that are of our interest.

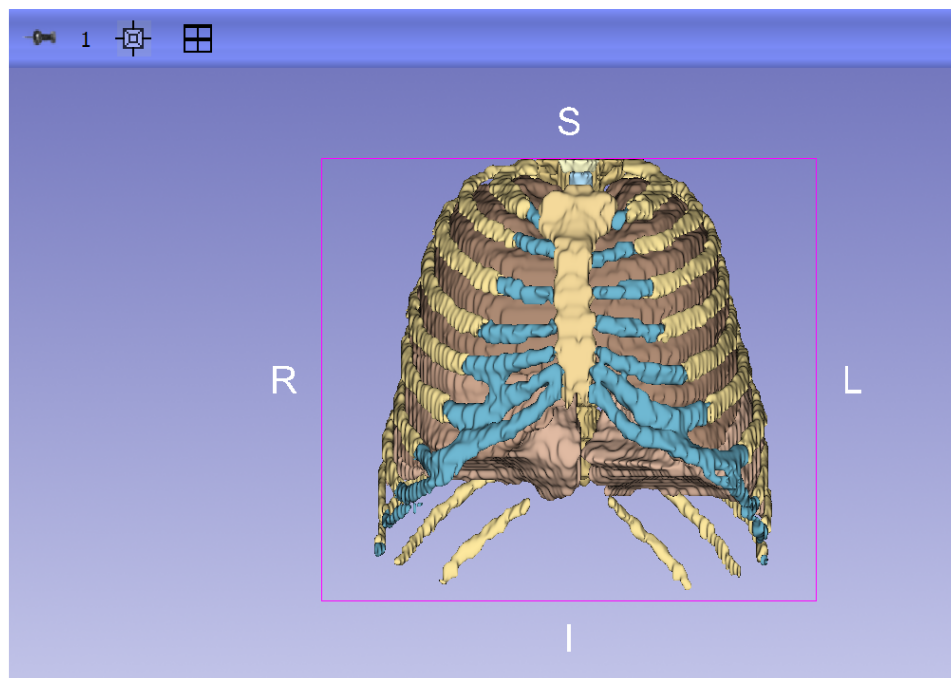


Figure 18: Final figure of the segmentation of the lungs and thoracic bones.

- Is it possible to use some IA module to obtain the same result?
- If not, which is the closer approximation that you can obtain automatically?
- From the obtained approximation, could it be possible to achieve the same result using the Segment Editor module?
- Explain how (it is not necessary to compute the segmentation, just only explain the procedure to follow).

So, finally yes, it is possible to use an IA module to obtain the same result that the obtained before, in fact if with this plugin we would haven't used the fast option in the menu, we would have obtained better results than now or better that the obtained by classical segmentation methods.

It is possible to obtain the same results using only the tools that provide the Segment Editor module because it is similar to the way that this plugin works, by creating precise segmentation segments. The drawback is that to do this work it would be necessary to use a lot of time on segmentation work instead of the one hour of work that the plugin needs to obtain a precise segmentation result.

The procedure to follow to obtain the same result using the segment editor module will be based on the tools explained on the first exercise: the threshold segmentation, the scissors segmentation and the island segmentation.

6 Plugin Task

Choose a plugin (not explained in the lab sessions) that allows you to work with the volume model chosen by providing advanced tools tailored to specific tasks.

The plugin or extension selected is the Airway segmentation. This is a simple module that segments the airway from a CT image. The user needs to specify the input volume and markup point placed in the trachea. The module automatically retrieves the convolution kernel of the image loaded from the DICOM.

To extract the segmentation view it has been done the next steps:

- The first step is to select the CT Chest volume data from the downloaded sample data.
- The second step is to create a point into a part of the airway volume that we want to segmentate. To do it in the markup module we click a point list and select the part in any vision of the volume. The point created is the F-1 that is placed into the trachea as it can be seen in the *Fig. 19*.

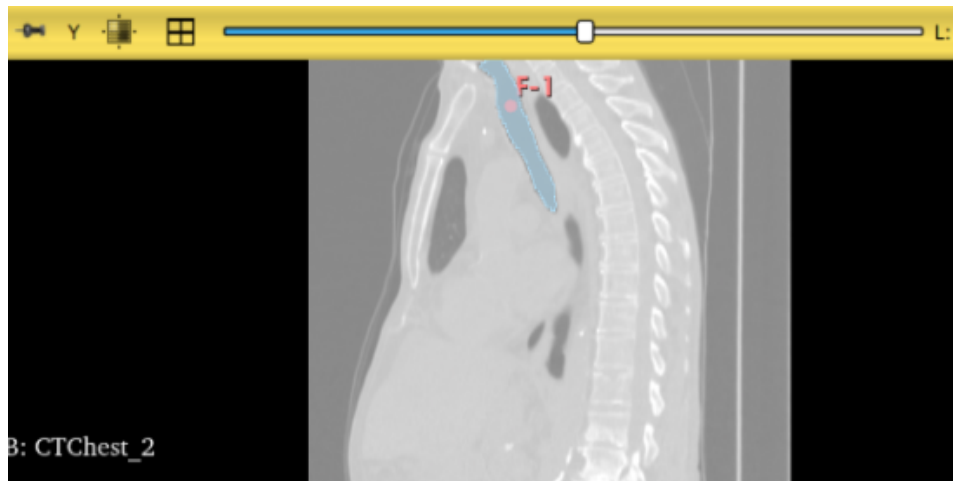


Figure 19: Visualization of the vertical and lateral vision of the chest and the point F-1.

- Finally in the Airway segmentation module it is selected the CT Chest volume and the point F-1 and it is applied to the segmentation. The F-1 point is clearly on the trachea of the patient since it is in the middle part of this vision of the chest. The air sections that can be seen in the *Fig. 20* correspond to the internal extreme parts of the right lung.

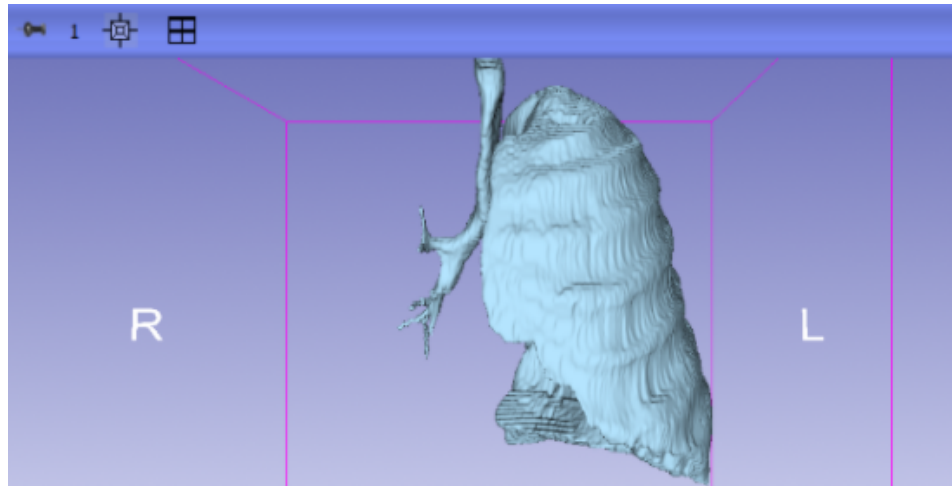


Figure 20: Final result of the Airway Segmentation AI tool on the CTChest data based on the marker point used.

In the *Fig. 20* can be observed as the result of the segmentation tool Airway Segmentation on the CTChest volume. It is clear how the segmentation tool shows the left lung but not the right. The correct side is clearly the right side because this segmentation extension was clearly to create images of air cavities as the trachea. The reason why this happens is because of a connection between the trachea and the left lung. The extension has a tool to determine the critical point of the trachea that connects to the lung - the bronchi - and uses it to differentiate the organs. For the right lung has properly worked since the terminations of the trachea that connects to the bronchi but for the left side it hasn't worked properly since the lung appears in the figure as an airway conduct: the trachea.

[@KitwareMedical/slicer-extensions-webapp](#)

[Update README.md](#) · [Slicer/SlicerAirwaySegmentation@d3954e4](#) · [GitHub](#)

7 Conclusions

The segmentation and visualization of the medical data is a basic tool needed for medical professionals to work. The data sensed in this area is full of different body parts but normally to analyse a condition is needed only to visualize a reduced number of organs. This is a complex work that needs to be simplified to properly do the diagnosis and treatment work of the medical professionals.

To achieve that goal, the program 3D slicer provides different tools to work with the medical data from different points of interest. In this work it has been seen different segmentation tools that the program offers, it has created different measurements that can help the medical professional work and has been created final images for a professional report.

Moreover has been tested different plugins to segmentate the chosen model based on processment technics as the artificial intelligence processment to simplify the segmentation work for different models. It has been proven the existence of a strong community of 3D slicer users that create different plugins to work with different models and objectives and share it worldwide.

The visualization of the medical data is a basic tool for the medical professionals, with the newest data acquisition systems created it is possible to accurately describe the human body of the patient but more important is to find the adequate information in all the data that will give the key point to know exactly the health state of the patient. That is why simpler and effective tools of segmentation are a central interest point in the medical application.